

Math 548
Portfolio Assignment 1

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Problem 1 (S24, P5). Let $T : \mathbf{R}^4 \rightarrow \mathbf{R}^4$ be an orthogonal projection to the 2-dimensional plane P spanned by the vectors $\vec{v} = (2, 0, 1, 0)$ and $\vec{w} = (-1, 0, 2, 0)$.

- (a) Find (with proof) all the eigenvalues and eigenvectors, along with their geometric and algebraic multiplicities.
- (b) Find the matrix T with respect to the standard basis. Is this matrix diagonalizable? Why or why not?

Proof. (a) Since T is a projection onto P , note that for any vector $v \in \mathbf{R}^4$, we have that $T(v) \in P$. Since T acts as the identity map on any vector in P , it must be the case that $T^2(v) = T(v)$.

Let w be an eigenvector of T . Since $T^2(v) = T(v)$ for all $v \in \mathbf{R}^4$, we get:

$$T(w) = \lambda w,$$

$$T(w) = T^2(w) = \lambda^2 w.$$

It must be the case that $\lambda^2 = \lambda$; i.e., $\lambda = 0$ or $\lambda = 1$. Note that $\lambda = 1$ corresponds to the subspace P , while $\lambda = 0$ corresponds to the subspace $P^\perp$. Since P is 2-dimensional, the geometric multiplicity of $\lambda = 1$ is 2. Moreover, because $\dim \mathbf{R}^4 = \dim P + \dim P^\perp$, we get that $\dim P^\perp = 2$, hence $\lambda = 0$ has a geometric multiplicity of 2.

Observe that:

$$\begin{aligned} \begin{pmatrix} 2 & 0 & 1 & 0 \\ -1 & 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} &= \begin{pmatrix} 0 \\ 0 \end{pmatrix} \implies \begin{cases} 2x_1 + x_3 = 0 \\ -x_1 + 2x_3 = 0 \end{cases} \\ &\implies \begin{cases} 2x_1 + x_3 = 0 \\ -5x_1 = 0 \end{cases} \\ &\implies \begin{cases} 2x_1 = -x_3 \\ x_1 = 0 \end{cases} \end{aligned}$$

Hence $x_1 = x_3 = 0$. Since x_2 and x_4 are free variables, we have:

$$P^\perp = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}.$$

Note that $P = E_1$ and $P^\perp = E_0$, where E_1 is the eigenspace associated to $\lambda = 1$ and E_0 is the eigenspace associated to $\lambda = 0$. Since $\dim(E_1) + \dim(E_0) = \dim(\mathbf{R}^4)$, T is diagonalizable. Because T is diagonalizable, the algebraic multiplicities of its eigenvalues equals their geometric multiplicities.

(b) Let $\mathcal{B}$ denote the standard basis in $\mathbf{R}^4$ and define:

$$S = \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Since T is diagonalizable, this means:

$$\begin{aligned} [T]_{\mathcal{B}} &= S \cdot [T]_{\mathcal{B}_\lambda} \cdot S^{-1} \\ &= \begin{pmatrix} 2 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{2}{5} & 0 & \frac{1}{5} & 0 \\ -\frac{1}{5} & 0 & \frac{2}{5} & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \end{aligned}$$

□

Problem 2 (S19, P1). Let L be the line in $\mathbf{R}^2$ defined by $y = -3x$, and let $T : \mathbf{R}^2 \rightarrow \mathbf{R}^2$ be the linear transformation that orthogonally projects onto L and then stretches along L by a factor of two.

- Find the eigenvalues and an eigenbasis $\mathcal{B}$ for T .
- Determine the matrix for T with respect to the basis $\mathcal{B}$.
- Determine the matrix for T with respect to the standard basis.

Proof. (a) Note that $\mathbf{R}^2 = L \oplus L^\perp$. Hence for any $v \in \mathbf{R}^2$, we can write $v = u + w$, where $u \in L$ and $w \in L^\perp$. This means:

$$\begin{aligned} T(v) &= T(u + w) \\ &= T(u) + T(w) \\ &= T(u) \\ &= 2u. \end{aligned}$$

Applying T to both sides yields:

$$\begin{aligned} T^2(v) &= 2T(u) \\ &= 2T(v). \end{aligned}$$

Whence $T^2 = 2T$. If x is an eigenvector of T , then $T(x) = \lambda x$ and $T(x) = \frac{1}{2}T^2(x) = \frac{1}{2}\lambda^2 x$. Thus $\lambda = \frac{1}{2}\lambda^2$, giving $\lambda = 2$ and $\lambda = 0$ as eigenvectors.

Note that $L = \text{span} \left\{ \begin{pmatrix} 1 \\ -3 \end{pmatrix} \right\}$. Solving the equation $(1, -3) \cdot \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = (0, 0)$ gives $L^\perp = \text{span} \left\{ \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right\}$. This means the eigenbasis for T is $\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ -3 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \end{pmatrix} \right\}$.

(b) Since $\dim \mathcal{B} = \dim \mathbf{R}^2$, the matrix for T will be diagonal. In particular, the diagonal entries for $[T]_{\mathcal{B}}$ will be the eigenvalues of T ; i.e., $[T]_{\mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}$.

(c) Let $\mathcal{E}$ denote the standard basis in $\mathbf{R}^2$. Since T is diagonalizable, we have $[T]_{\mathcal{B}} = P^{-1} \cdot [T]_{\mathcal{E}} \cdot P$, where $P = \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix}$. Solving for $[T]_{\mathcal{E}}$ yields:

$$\begin{aligned} [T]_{\mathcal{E}} &= P \cdot [T]_{\mathcal{B}} \cdot P^{-1} \\ &= \begin{pmatrix} 1 & 3 \\ -3 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{10} & -\frac{3}{10} \\ \frac{3}{10} & \frac{1}{10} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{5} & -\frac{3}{5} \\ -\frac{3}{5} & \frac{1}{5} \end{pmatrix}. \end{aligned}$$

□

Problem 3 (S21, P4). Let $A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$.

- Determine whether A is diagonalizable, and if so, give its diagonal form along with a diagonalizing matrix.
- Compute A^{42} . Remember to show all work.

Proof. (a) Observe that:

$$\begin{aligned} 0 &= \det(A - \lambda I) \\ &= \det \begin{pmatrix} -\lambda & 0 & -2 \\ 1 & 2-\lambda & 1 \\ 1 & 0 & 3-\lambda \end{pmatrix} \\ &= -\lambda \cdot \det \begin{pmatrix} 2-\lambda & 1 \\ 0 & 3-\lambda \end{pmatrix} - 2 \det \begin{pmatrix} 1 & 2-\lambda \\ 1 & 0 \end{pmatrix} \\ &= -\lambda(2-\lambda)(3-\lambda) - 2(\lambda-2) \\ &= -(\lambda-2)^2(\lambda-1) \end{aligned}$$

Thus $\lambda = 2$ has an algebraic multiplicity of 2, while $\lambda = 1$ has an algebraic multiplicity of 1. Now consider:

$$\begin{aligned} (A - 2I)\vec{x} &= \vec{0} \implies \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{cases} x_1 + x_3 = 0 \end{cases} \end{aligned}$$

We know x_2 is free, so let x_3 also be free. This gives $\mathcal{B}_{E_2} = \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$ as a basis for E_2 .

Similarly:

$$\begin{aligned} (A - I)\vec{x} = \vec{0} &\implies \begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\begin{cases} x_2 = x_3 \\ x_1 = -2x_3 \end{cases} \end{aligned}$$

Since x_3 is free, this gives $\mathcal{B}_{E_1} = \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$ as a basis for E_1 . Since $\dim(E_1) + \dim(E_2) = \dim(\mathbf{R}^3)$, A is diagonalizable. In particular, $A = PDP^{-1}$, where:

$$P = \begin{pmatrix} -2 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

(b) Since $A = PDP^{-1}$, we have:

$$\begin{aligned} A^{42} &= PD^{42}P^{-1} \\ &= \begin{pmatrix} -2 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2^{42} & 0 & 0 \\ 0 & 2^{42} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -2 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}^{-1} \\ &= \dots \end{aligned}$$

□

Problem 4 (S22, P5). Let $A = \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix}$.

- Compute the characteristic polynomial $p_A(x)$ of A . It has integer roots.
- For each eigenvalue λ of A , find a basis for the eigenspace E_λ .
- Determine if A is diagonalizable. If so, give matrices P and B such that $P^{-1}AP = B$ and B is diagonal. If not, explain carefully why A is not diagonalizable.

Proof. (a) Observe that:

$$\begin{aligned}
 0 &= \det(A - \lambda I) \\
 &= \det \begin{pmatrix} 2-\lambda & -1 & -1 \\ 1 & -\lambda & -1 \\ 1 & -1 & -\lambda \end{pmatrix} \\
 &= \det \begin{pmatrix} -1 & -1 \\ -\lambda & -1 \end{pmatrix} + \det \begin{pmatrix} 2-\lambda & -1 \\ 1 & -1 \end{pmatrix} - \lambda \det \begin{pmatrix} 2-\lambda & -1 \\ 1 & -\lambda \end{pmatrix} \\
 &= (1-\lambda) + (\lambda-2+1) - \lambda(-2\lambda + \lambda^2 + 1) \\
 &= -\lambda(\lambda-1)^2.
 \end{aligned}$$

Hence the eigenvalues of A are $\lambda = 0$ with an algebraic multiplicity of 1, and $\lambda = 1$ with an algebraic multiplicity of 2.

(b) For $\lambda = 0$:

$$\begin{aligned}
 A\vec{x} = \vec{0} &\implies \begin{pmatrix} 2 & -1 & -1 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\implies \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\implies \begin{pmatrix} 1 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\implies \begin{cases} x_1 = x_2 \\ x_1 = x_3 \end{cases}
 \end{aligned}$$

Take x_1 to be a free variable. We have $\mathcal{B}_{E_0} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$ as a basis for E_0 .

Similarly for $\lambda = 1$:

$$\begin{aligned}
 (A - I)\vec{x} = \vec{0} &\implies \begin{pmatrix} 1 & -1 & -1 \\ 1 & -1 & -1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\implies \begin{pmatrix} 1 & -1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
 &\implies x_1 = x_2 + x_3.
 \end{aligned}$$

Since x_2 and x_3 are free variables, we have $\mathcal{B}_{E_1} = \left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ as a basis for E_1 .

(c) Since $\dim(E_1) + \dim(E_0) = \dim(\mathbf{R}^3)$, A is diagonalizable. Let:

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Then $P^{-1}AP = B$. □

Problem 5. Let $A = \begin{pmatrix} 6 & -2 & -1 \\ 10 & -3 & -2 \\ 0 & 0 & 1 \end{pmatrix}$.

- (a) Find bases for the eigenspaces of A .
- (b) Determine if A is diagonalizable. If so, give an invertible matrix P and a diagonal matrix D such that $P^{-1}AP = D$. If not, explain why not.

Proof. (a) Observe that:

$$\begin{aligned} 0 &= \det(A - \lambda I) \\ &= \det \begin{pmatrix} 6-\lambda & -2 & -1 \\ 10 & -3-\lambda & -2 \\ 0 & 0 & 1-\lambda \end{pmatrix} \\ &= -(\lambda - 1) \det \begin{pmatrix} 6-\lambda & -2 \\ 10 & -3-\lambda \end{pmatrix} \\ &= -(\lambda - 1)(\lambda - 6)(\lambda + 3) + 20 \\ &= -(\lambda - 1)(\lambda^2 - 3\lambda + 2) \\ &= -(\lambda - 1)^2(\lambda - 2). \end{aligned}$$

Hence $\lambda = 1$ and $\lambda = 2$ are eigenvalues of A .

For $\lambda = 1$:

$$\begin{aligned} (A - I)\vec{x} &= \vec{0} \implies \begin{pmatrix} 5 & -2 & -1 \\ 10 & -4 & -2 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{pmatrix} 5 & -2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies 5x_1 = 2x_2 + x_3 \end{aligned}$$

Since x_2 and x_3 are free variables, we have $\mathcal{B}_{E_1} = \left\{ \begin{pmatrix} \frac{2}{5} \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{5} \\ 0 \\ 1 \end{pmatrix} \right\}$ as a basis for E_1 .

For $\lambda = 2$:

$$\begin{aligned}(A - 2I)\vec{x} = \vec{0} &\implies \begin{pmatrix} 4 & -2 & -1 \\ 10 & -5 & -2 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{pmatrix} 2 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\ &\implies \begin{cases} 2x_1 = x_2 \\ x_3 = 0 \end{cases}\end{aligned}$$

Since x_2 is a free variable, we have $\mathcal{B}_{E_2} = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \right\}$.

(b) Since $\dim(E_1) + \dim(E_2) = \dim(\mathbf{R}^3)$, A is diagonalizable. In particular, we have:

$$P = \begin{pmatrix} 2 & 1 & 2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix},$$

where $P^{-1}AP = D$.

□